INCH-POUND

MIL-PRF-49471/10A(CR)
30 November 2000
Superseding
MIL-PRF-49471/10(ER)
2 June 1995

PERFORMANCE SPECIFICATION SHEET
BATTERY, NON-RECHARGEABLE, HIGH PERFORMANCE, WATERTIGHT, BA-X112/U

This specification is approved for use within Communications Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification and MIL-PRF-49471(CR).

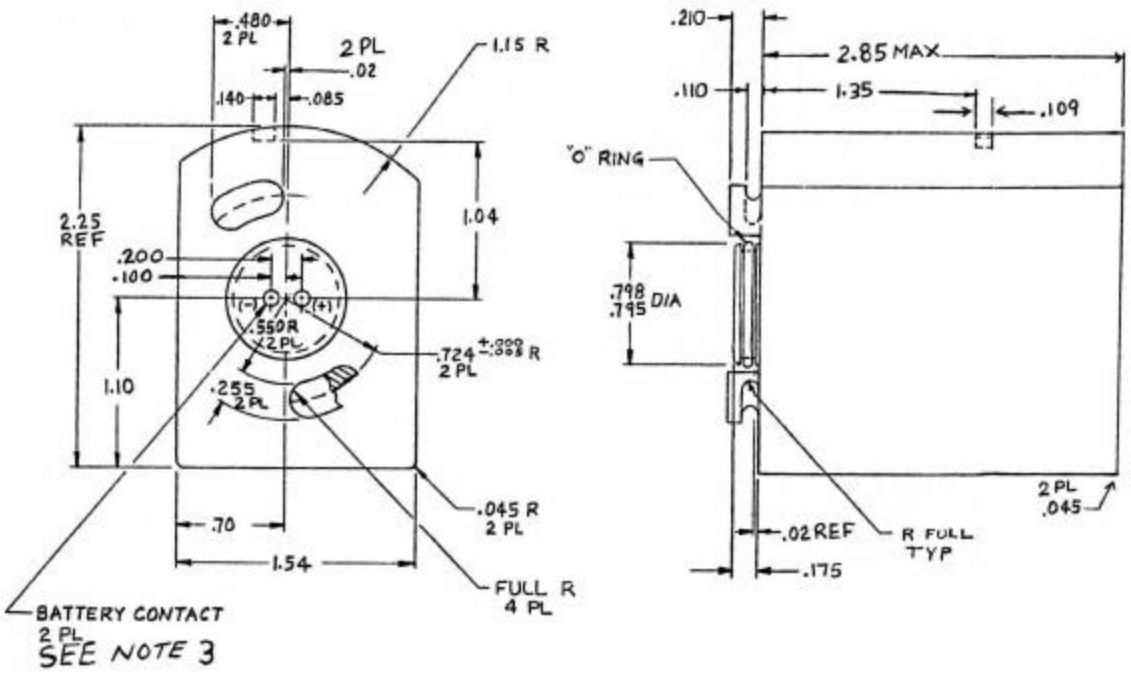


Figure 1. Battery dimensions.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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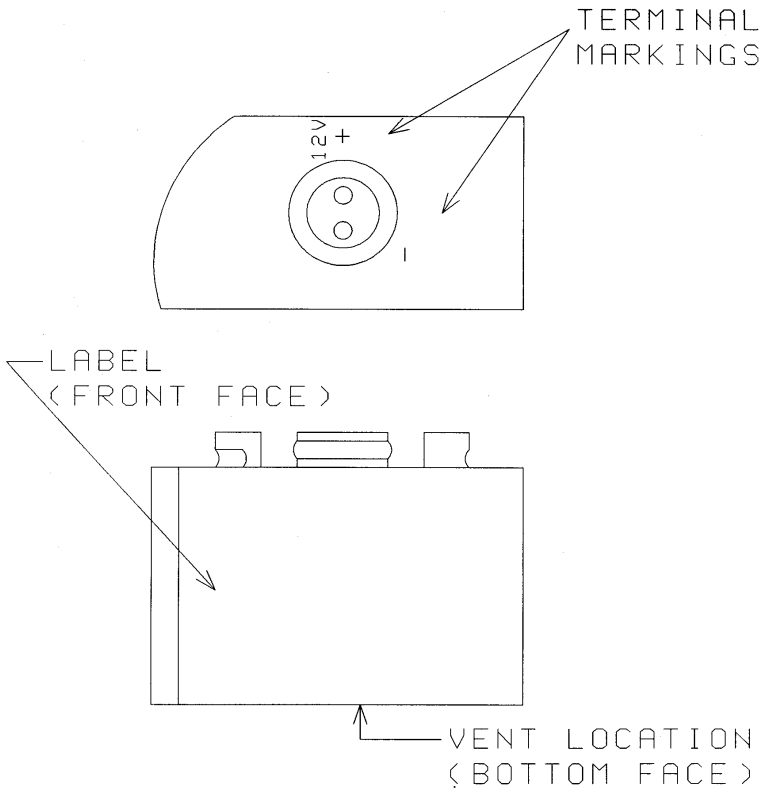


Figure 2. Terminal Markings

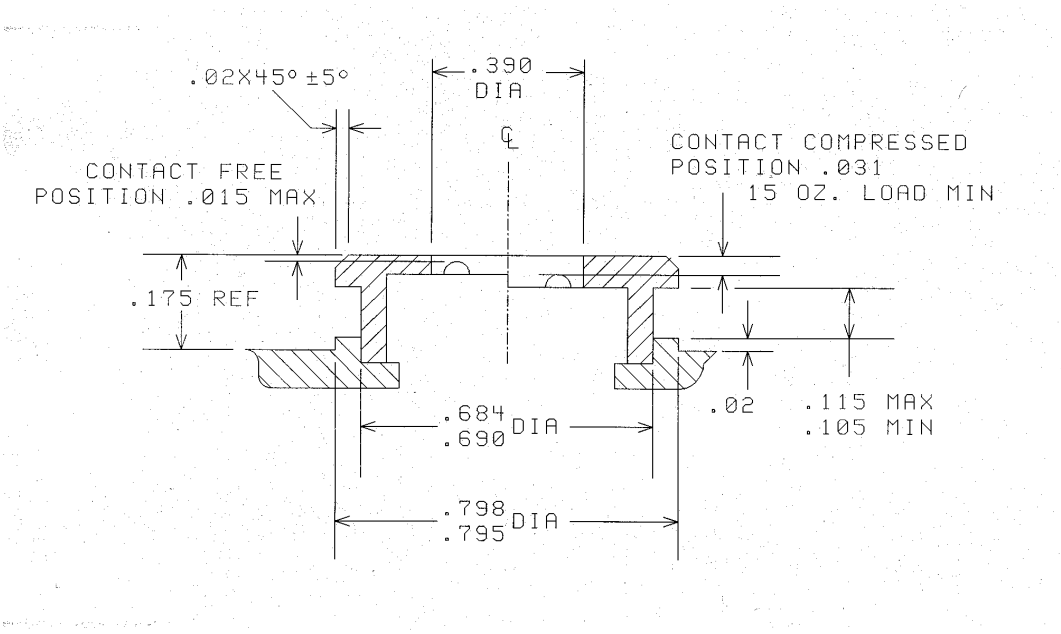


Figure 3. Contact Configuration

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NOTES:

1. All dimensions are in inches.

2. Tolerances are: 2 place decimals ±.01. 3 place decimals ±.005.

3. The contacts shall not protrude above the surface of the battery terminal. The contact's diameter shall be .109±.005 inch and the contact surface shall be smooth.

REQUIREMENTS:

Dimensions and configuration: See Figure 1 and Figure 3.

Maximum voltage: 13.20 volts.

Cut-off voltage for capacity: 8.00 volts.

Terminals: 2 pin contact, coupler interface, "O" ring sealed (see Figure 1 and 3).

Weight (maximum): 0.60 pounds (272 gms).

Charge protection: Applicable.

Battery short circuit test: Applicable.

High temperature protection: Applicable.

Complete discharge device: Applicable.

State of Charge Indicator: Not applicable.

Battery contacts: The contacts shall be constructed from tubular, oval head rivets conforming to MS16535-11 per NASM16535. The contacts shall be nickel plated, 0.0010 inch minimum thickness. Then the contacts shall be gold plated, 0.00010 inch minimum thickness, approximately 0.0456 square inches. The contacts shall be mounted on a spring-loaded surface. The contacts shall require a minimum mating force of 15 ounces per contact.

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Capacity tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

Capacity Test	Battery Capacity Requirements in Hours to 8.00 Volts
I	3.00
L	1.50
H	5.50
HT	5.30
LT	1.30
IT	2.80
LR1	45.0

Initial voltage delay: During the I, H, HT, and IT tests, initial voltages below the 8.00 volt minimum voltage cannot exceed a 5.0 second duration. During L and LT discharge tests initial voltage below 8.00 volts cannot exceed a 60.0 second duration.

Venting mechanism: The battery case shall incorporate a venting mechanism that shall prevent the battery case from exploding or rupturing into fragments when cell venting occurs. The venting mechanism shall be located at the bottom face (see Figure 2). Venting mechanism shall not become separated from the battery case when activated by cell venting.

Watertight Integrity: The battery, when subjected to the Watertight Integrity test, shall not increase in weight by more than 0.1 grams and no water shall come into contact with the battery terminals. There shall be no sign of corrosion on the battery contacts.

METHOD OF EXAMINATION AND TEST:

Capacity tests:

(1) Storage: Details on storage conditions for all specified capacity tests are described in the basic specification.

(2) Discharge:

Cells (forced discharge): One cell shall be discharged at 0.65 amperes at 70 ± 5°F to 2.0 volts. The discharged cell shall then be connected in series to three fresh cells. Each string shall be discharged at 0.65 amperes at 70 ± 5°F for 3.0 hours.

Batteries: For all capacity tests, except the LR1, H and HT, the batteries shall be discharged at 0.65 amperes constant current to cut-off volts (zero volts for the I test). For the H and HT capacity tests the batteries shall be discharged through 0.65 amperes constant current for five minutes followed by open-circuit for five minutes. This cycle shall be repeated until the closed-circuit voltage reaches cut-off volts. For the LR1 test, batteries shall be discharged at 45 milliamperes to 8.5 volts, then at 0.7 watts to 1 volt.

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Forced discharge: After discharge to zero volts on the I capacity test, batteries shall be discharged at 0.65 amperes for a minimum of 5 minutes.

Overload: Batteries shall be discharged at 1.3 amperes.

Abuse test, predischarge: Designated samples shall be discharged at 0.65 amperes for 1.5 hours.

Abuse test, pulse discharge: Batteries shall be discharged at 0.88 amperes for 1 minute followed by 0.64 amperes for 4 minutes, cycled continuously to cut-off voltage.

Closed-circuit voltage (battery & cell string): Load resistance of 12.25 ohms shall be used between battery terminals. Voltage shall be above 8.0 volts within 10 seconds.

Cell closed circuit voltage: Load resistance of 3 ohms shall be used. Voltage shall be above 2.0 volts within 5 seconds.

Venting mechanism: The battery case venting mechanism shall be monitored during the safety feature test. During cell venting the battery venting mechanism shall release the pressure build up in the battery case and shall meet the performance requirements specified herein.

Watertight Integrity Test: The First Article Group II samples and the Group C quality conformance inspection samples shall be subject to the watertight integrity test after environmental testing and capacity testing. After capacity testing, the weight of each sample shall be measured and recorded. Each sample shall be mated to a Motorola part number 64-P21451J001 coupler plate or equal. The coupler shall be waterproofed, except at the battery/coupler plate interface, to prevent water from contacting the battery connector contacts. The battery/plate assemblies shall be immersed in a synthetic solution containing 3.6 percent by weight sea salt in accordance with ASTM-D-1141. The solution with the battery samples shall be pressurized to simulate two foot immersion for 24 hours and the pressure shall be raised to simulate 50 foot immersion for 15 minutes. The battery samples shall be towel dried and left to air dry for 15 minutes. The batteries shall be weighed and shall meet the watertight integrity seal requirements specified.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army - CR

Preparing Activity:
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(Project 6135-0417-10)

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